

Benchmarking Data-Integrity Anomaly Detection in Blood-Bank Transaction Records

Aryan Jain
High School Graduate
India

Abstract—Blood Establishment Computer Systems (BECS) operate under strict regulatory frameworks to ensure the traceability of the blood supply. While unsupervised machine learning (ML) is increasingly used for healthcare anomaly detection, the suitability of purely data-driven anomaly detection for discrete, domain-specific blood-bank integrity constraints remains less explored than its use for general statistical outlier detection. This paper evaluates a hybrid framework combining a deterministic rule engine with a Median Absolute Deviation (MAD) statistical detector against out-of-sample Isolation Forest (IF) and Local Outlier Factor (LOF) ML baselines. Using a synthetic taxonomy of 300 mutually exclusive anomalies injected across 2,000 records, evaluated over 10 independent train-test seed pairs, the hybrid framework achieved complete coverage of the predefined benchmark. In contrast, the tested ML baselines were inconsistent across anomaly classes (IF F1: 0.313 ± 0.024 ; LOF F1: 0.370 ± 0.041). These results demonstrate the potential complementarity of deterministic validation and robust statistical detection for maintaining data integrity in highly regulated clinical environments.

Index Terms—Data mining, anomaly detection, blood bank, health information systems, synthetic data.

I. INTRODUCTION

Modern blood banks rely on Blood Establishment Computer Systems (BECS) to systematically mitigate risks and maintain data integrity. BECS platforms must enforce validated workflows and time-stamped audit trails to maintain a reliable chain of custody [1].

Data mining techniques are frequently deployed to audit healthcare information systems, identifying transactional anomalies indicative of human error or software glitches [7]. Unsupervised machine learning (ML) models, such as Isolation Forests, are often favored due to their ability to detect novel patterns in high-dimensional data [4]. However, ML models optimize for statistical variance and spatial distancing, which may limit their capacity to capture discrete regulatory violations, such as an issued unit lacking a valid donor reference.

This research frames its contributions around three primary areas:

- 1) A controlled synthetic benchmark containing distinct blood-bank data-integrity anomaly classes.
- 2) A robust out-of-sample comparison of deterministic and statistical detection, alongside IF and LOF baselines, evaluated by anomaly class over multiple independent seed generations.

- 3) A hybrid design implication demonstrating how rule-based and statistical methods can complement each other within a blood-bank validation architecture.

II. REGULATORY FRAMEWORK AND TRACEABILITY

The architectural requirements for data integrity in blood banking are dictated by regulatory mandates. WHO guidance emphasizes the importance of traceability, documentation, and controlled records in blood establishments [1]. Traceability requires systems capable of forward and backward tracking, linking the donor's intake profile directly to final disposition [8].

In the Indian clinical context, biological tracking requirements are codified by the Drugs and Cosmetics Act of 1940 and its subsequent Rules [2]. Blood centers must maintain exhaustive records for processing human blood, ensuring the chain of custody remains unbroken. Global systems often implement standards like ISBT 128 to provide international consistency for barcode labeling and traceability [3].

From a data-mining perspective, ensuring compliance requires verifying that clinical data conforms to a complex state machine. For the purposes of this benchmark, the blood-unit lifecycle is represented conceptually as

COLLECTED $\rightarrow$ TESTED $\rightarrow$ AVAILABLE $\rightarrow$ ISSUED.

Invalid transitions, such as COLLECTED $\rightarrow$ ISSUED (bypassing mandatory testing), are anomalous structural constraints that systems must enforce. Similarly, an issued unit with a reactive test result represents an invalid lifecycle condition in the synthetic benchmark.

III. ALGORITHMIC FRAMEWORK AND DATA ENGINEERING

Due to the sensitive nature of protected health information, a synthetic dataset with known ground truth was generated to facilitate reproducible benchmarking [6].

A. Data Synthesis and Population Modeling

The engineering process initialized a registry of 500 synthetic donors. Blood-group assignments were generated using fixed probabilities chosen to produce a heterogeneous synthetic donor population. Using this registry, 2,000 unit transactions were simulated over a 90-day operational window. The units were partitioned into four clinical components with specific regulatory shelf lives: Packed Red Blood Cells (42 days), Whole Blood (35 days), Fresh Frozen Plasma (365

days), and Platelets (5 days). Clean data generation mathematically guarantees zero internal constraints are violated prior to injection.

B. Controlled Error Taxonomy

To establish a rigorous mathematical benchmark, exactly 300 anomalies (15% anomaly ratio) were injected into the synthetic dataset. The injected anomalies were mutually exclusive at the transaction level, such that each anomalous transaction was assigned to exactly one taxonomic class. Duplicate injection mechanisms were strictly constrained to copy from verified clean records to preserve this mutual exclusivity. This taxonomy encompasses nine distinct failure modes:

- **Missing Fields (n=34):** Mandatory data points arbitrarily removed.
- **Duplicate Records (n=34):** Unit identifier collisions.
- **Conflicting Records (n=34):** Unit blood group contradicts the donor’s historical data, simulating a transaction-level entry error.
- **Impossible States (n=33):** Units issued despite reactive/discard tests.
- **Expiry Violations (n=33):** Issue timestamps succeeding the biological expiration limit.
- **Traceability Breaks (n=33):** Transaction references a donor identifier that cannot be resolved against the registry.
- **Format Inconsistencies (n=33):** Free-text corruptions of constrained enumerations.
- **Temporal Anomalies (n=33):** Issue timestamps chronologically prior to collection timestamps.
- **Extreme Univariate Numerical Deviations (n=33):** Extreme synthetic weight (188.5 kg) and hemoglobin (24.8 g/dL) outliers.

C. Algorithmic Implementations

The benchmark evaluates three detection approaches and a hybrid integration.

Rule-Based Validator: A deterministic domain-integrity validator programmed with strict operational logic. It parses each transaction against verification layers: completeness, domain integrity, identifier uniqueness, referential integrity, issued-state constraints, and temporal logic.

Statistical Anomaly Detector: To capture numerical outliers, a statistical module utilized the Modified Z-Score based on the Median Absolute Deviation (MAD) [5], calibrated exclusively on the training set. Observations yielding a modified Z-score > 3.5 for continuous numerical variables on the test set were flagged as anomalous.

Machine Learning Baselines (IF and LOF): An Isolation Forest [4] and a Local Outlier Factor (LOF) model were deployed as unsupervised feature-space anomaly-detection baselines with contamination $c = 0.15$. Input features comprised donor weight, donor hemoglobin, quantity issued, collection-to-issue lead time, and one-hot encoded categorical variables. To prevent feature-space mismatch between independent splits, a global OneHotEncoder was fitted to a predefined vocabulary.

TABLE I
OVERALL OUT-OF-SAMPLE BENCHMARK RESULTS (MEAN $\pm$ SD)

Method	Precision	Recall	F1	FPR
Rule	1.000 $\pm$ 0.000	0.890 $\pm$ 0.000	0.942 $\pm$ 0.000	0.000 $\pm$ 0.000
MAD	1.000 $\pm$ 0.000	0.110 $\pm$ 0.000	0.198 $\pm$ 0.000	0.000 $\pm$ 0.000
IF	0.310 $\pm$ 0.029	0.316 $\pm$ 0.021	0.313 $\pm$ 0.024	0.125 $\pm$ 0.011
LOF	0.351 $\pm$ 0.046	0.394 $\pm$ 0.045	0.370 $\pm$ 0.041	0.130 $\pm$ 0.022
Hybrid	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000

TABLE II
SENSITIVITY ANALYSIS: ISOLATION FOREST (OOS)

Contam.	Recall	FPR	F1
0.05	0.120 $\pm$ 0.028	0.040 $\pm$ 0.008	0.178 $\pm$ 0.042
0.10	0.268 $\pm$ 0.026	0.077 $\pm$ 0.009	0.314 $\pm$ 0.032
0.15	0.316 $\pm$ 0.020	0.125 $\pm$ 0.010	0.313 $\pm$ 0.023
0.20	0.354 $\pm$ 0.022	0.175 $\pm$ 0.011	0.302 $\pm$ 0.020

Missing fields and malformed text formats were explicitly cast as distinct categorical states.

Hybrid Detector: The final anomaly decision was defined as the logical OR of the rule-based and MAD detector outputs. A transaction was classified as anomalous if either deterministic validation or MAD-based outlier detection flagged it.

To ensure generalizability, 10 independent train and test seed pairs were generated. ML models and MAD parameters were strictly fitted on the training datasets and evaluated out-of-sample (OOS) on the independent test datasets. Experiments were executed using Python 3.12.1, NumPy 1.26.4, pandas 2.2.0, and scikit-learn 1.4.2 to facilitate reproducibility.

IV. EMPIRICAL BENCHMARKING AND RESULTS

Table I presents the overall out-of-sample classification performance averaged across the 10 independent seed pairs.

The Rule-Based Validator achieved perfect precision on deterministic constraints but predictably missed the numerical outliers. The MAD detector successfully isolated the extreme univariate numerical deviations but bypassed structural violations. The Hybrid model successfully captured all 300 predefined anomalies.

Because unsupervised ML performance is highly sensitive to defined parameters, Table II details an IF sensitivity analysis evaluated out-of-sample. Using $c = 0.15$ utilizes knowledge of the benchmark’s injected anomaly proportion; evaluating across thresholds demonstrates the necessary trade-offs between recall and false-positive rates.

An independent sensitivity analysis of the MAD detector’s magnitude parameters revealed that while it achieved 100.0% burst recall on *extreme* anomalies (188.5 kg, 24.8 g/dL), its recall dropped to 0.0% for *moderate* (115.0 kg) and *subtle* (98.0 kg) deviations.

Table III details algorithm recall across the taxonomic classes. Both unsupervised feature-space anomaly-detection baselines (IF and LOF) were inconsistent; they reliably captured numerical bursts and impossible states, but were sub-

TABLE III
PER-CLASS RECALL PERCENTAGES (MEAN $\pm$ SD, %)

Anomaly Class	Rule	MAD	IF	LOF
Missing Fields	100.0 $\pm$ 0.0	0.0 $\pm$ 0.0	27.6 $\pm$ 10.5	19.1 $\pm$ 6.6
Duplicate Records	100.0 $\pm$ 0.0	0.0 $\pm$ 0.0	12.6 $\pm$ 3.2	11.2 $\pm$ 4.3
Conflicting Records	100.0 $\pm$ 0.0	0.0 $\pm$ 0.0	14.7 $\pm$ 5.6	11.2 $\pm$ 4.1
Impossible States	100.0 $\pm$ 0.0	0.0 $\pm$ 0.0	100.0 $\pm$ 0.0	77.6 $\pm$ 11.4
Expiry Violations	100.0 $\pm$ 0.0	0.0 $\pm$ 0.0	7.3 $\pm$ 5.9	42.7 $\pm$ 23.8
Traceability Breaks	100.0 $\pm$ 0.0	0.0 $\pm$ 0.0	7.3 $\pm$ 8.3	14.8 $\pm$ 5.2
Format Inconsistencies	100.0 $\pm$ 0.0	0.0 $\pm$ 0.0	16.1 $\pm$ 4.3	12.1 $\pm$ 4.5
Temporal Anomalies	100.0 $\pm$ 0.0	0.0 $\pm$ 0.0	3.3 $\pm$ 3.9	69.7 $\pm$ 14.1
Extreme Univariate	0.0 $\pm$ 0.0	100.0 $\pm$ 0.0	97.0 $\pm$ 9.1	98.5 $\pm$ 3.6

stantially weaker on several discrete integrity constraints than the deterministic validator. LOF outperformed IF on expiry violations (42.7% vs. 7.3%) and temporal anomalies (69.7% vs. 3.3%).

V. DISCUSSION AND LIMITATIONS

The empirical results reveal that different anomaly classes exhibit distinct mathematical structures. Deterministic constraints are naturally captured by rules, while extreme univariate numerical deviations can be captured by robust statistical methods. The tested ML configurations struggled because structural violations, such as unresolved donor identifiers, are not naturally represented as large numerical deviations in the feature space used by IF and LOF.

A false-positive analysis indicated that unsupervised spatial trees conflate statistical rarity with data invalidity. In a sample split, IF generated 234 false positives out of 1,700 clean records. While valid rare clinical states (like AB+ blood) represented only 6.9% of the clean population, they constituted 10.7% of the false-positive flags. Similarly, A+ blood composed 26.2% of clean records but 38.0% of false positives. A high false-positive rate could significantly increase the number of alerts requiring human review in a clinical setting.

It is crucial to acknowledge the limitations of this experimental design. The dataset and anomalies were synthetically generated and explicitly engineered to trigger the deterministic and statistical rules. The MAD detector successfully identified the numerical anomalies because they were instantiated as massive univariate outliers; its ability to detect subtle multivariate drift remains untested by this thresholding approach. Because the injected taxonomy was deliberately partitioned into deterministic structural errors and extreme numerical deviations, the perfect hybrid result measures coverage of the predefined benchmark rather than general-world anomaly-detection performance.

VI. CONCLUSIONS

The benchmark demonstrates complementary coverage of predefined structural and extreme numerical anomaly classes by deterministic and robust statistical detectors, evaluated out-of-sample across 10 independent datasets. The tested spatial-density ML baselines showed inconsistent class-level performance, highlighting the difference between statistical rarity and domain invalidity. The results motivate further evaluation

of hybrid validation architectures using independently generated and real-world datasets.

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